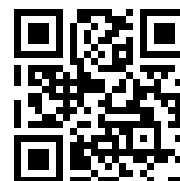


Evacuation Model

Explore the project

evacuate.mtbacheloma.org



Why this work matters

When many households leave at once, narrow roads and busy merges can create long queues. This project explores how traffic could move through our shared road network during an evacuation, helping our community identify bottlenecks, quantify risk, and test solutions.

How the model works

A computer simulation moves estimated household vehicles through connected roads and intersections. Departure timing, vehicle count, and evacuation route information are taken from our [Evacuation Modeling Survey](#). The simulation models each vehicle as it traverses its own evacuation route, considering stop and merge rules, single-lane constraints, and priority for emergency vehicles.

Your local knowledge makes the model better

Share how many vehicles your household would use, how long you would need to get ready, your primary and secondary routes, and any road or access issues you know about via our [Evacuation Modeling Survey](#)



What the model shows

- **Estimated evacuation times:** How long modeled traffic takes to clear the road network.
- **Traffic bottlenecks:** Where queues build up and constrain traffic flow.
- **Departure timing:** How the spread of household departures changes evacuation and commute times.

online only

 [View the Presentation](#)

Evacuation Modeling

Firewise Communities
Along Summit Corridor East of Highway 17

Amy Winkler, Firewise Leader | 2026 September 13

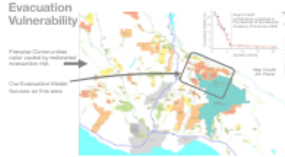
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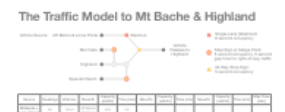
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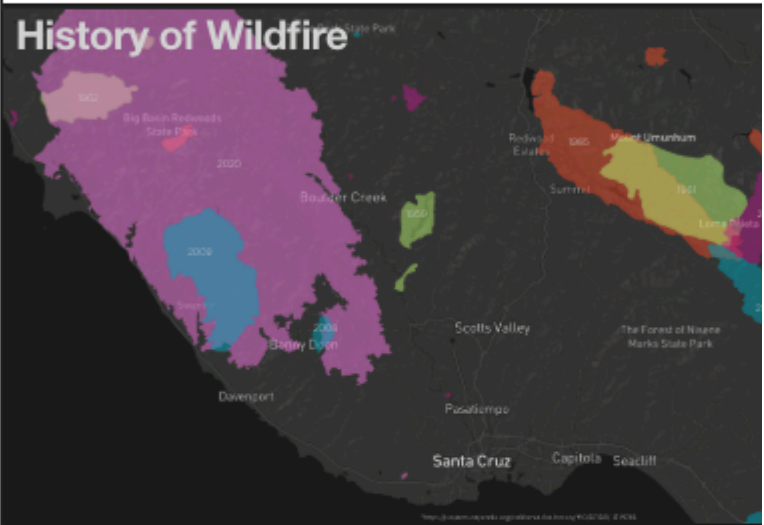
6



Evacuation Modeling

Firewise Communities
Along Summit Corridor East of Highway 17

Amy Winkler, Firewise Leader | 2026 September 13



History of Wildfire

Slide Credit:



PARADISE

85 deaths; 153,336 acres
Heavy traffic + rapidly deteriorating fire conditions produced widespread gridlock. NIST documented at least 19 burnovers.
Roads can become traps.



LAHAINA

102 deaths; 2,170 acres
Rapid fire, infrastructure failures and constrained evacuation left residents with little time. Older adults were disproportionately vulnerable.
Minutes matter.



CZU

1 death; 86,5
1,490 structures d
one civilian fatality experienced a m
evacuation — but i
than Paradise c
It can happen

Evacuation Vulnerability

